Variable Selection and Regularized Methods [1]

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[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

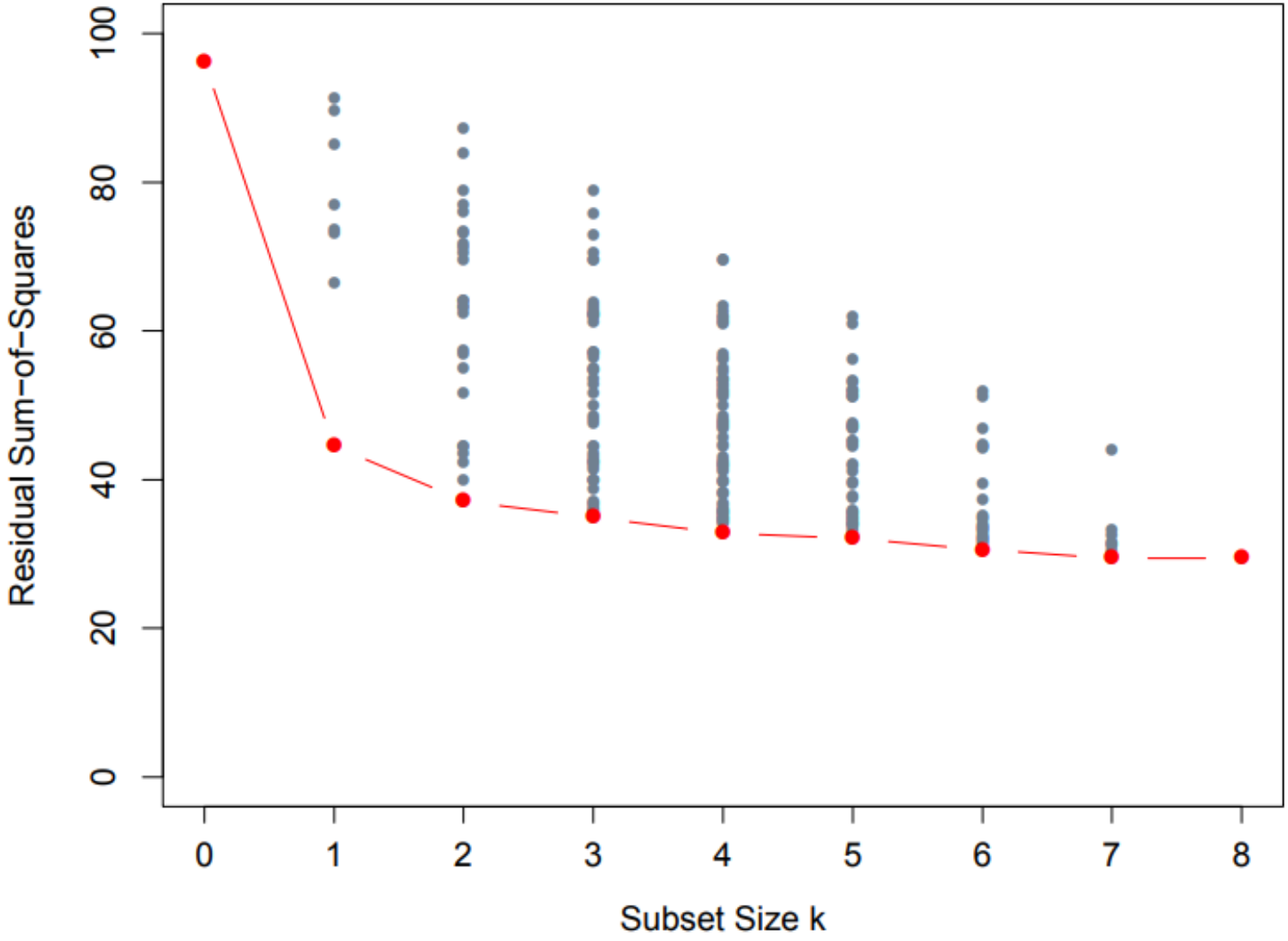
Variable Selection Methods For Linear Regression

- Why? Try to improve the model: exclude unnecessary predictors
  - Interpretation and simplicity
  - Prediction stability and accuracy
  - Less computational cost
  - Bias-variance tradeoff
- Common approaches
  - Subset selection
  - Shrinkage (also called *regularization*)
  - Dimension reduction (project p predictors to an M-dimensional subspace)
- Some times it is subjective, and needs domain knowledge that certain variables must be in the model.

Best Subset Selection

- Select the best subset of predictors such that the model is optimal in terms of a certain assessment metric
- Computationally expensive even infeasible
  - *leaps and bounds* (an R package "leaps") algorithm makes it feasible for p as large as 30 or 40.
- Suppose there are 10 predictors. How many models need to be fitted and evaluated?

Example of Best Subset Selection



Forward, Backward, and Stepwise Selection

- Computationally less expensive than best subset
- Iteratively adding or dropping one variable at a time
- Forward/backward is **greedy** procedure. That is, they won't adjust any added/dropped variables in previous step
- Stepwise: start with forward, and then iteratively add and drop variables
- Selection criteria: AIC or BIC
- R package: `"step"`
- An illustration: click here

Shrinkage Methods

- Also called penalized estimation, or regularization.
- Shrink the regression coefficients toward 0 by constraints (regularization)
- Estimates are usually *biased*
- Shrinkage methods are always preferred over best subset or stepwise methods. Why?
- A game of bias-variance tradeoff
- We discuss two popular shrinkage methods:
  - Ridge regression
  - LASSO

LASSO

- Least absolute shrinkage and selection operator (LASSO)
- Introduced by Tibshirani (1996)
- Shrinkage estimation
- It estimates the coefficients and perform variable selection simultaneously

LASSO

- LASSO solves the *(L_1) penalized least square*

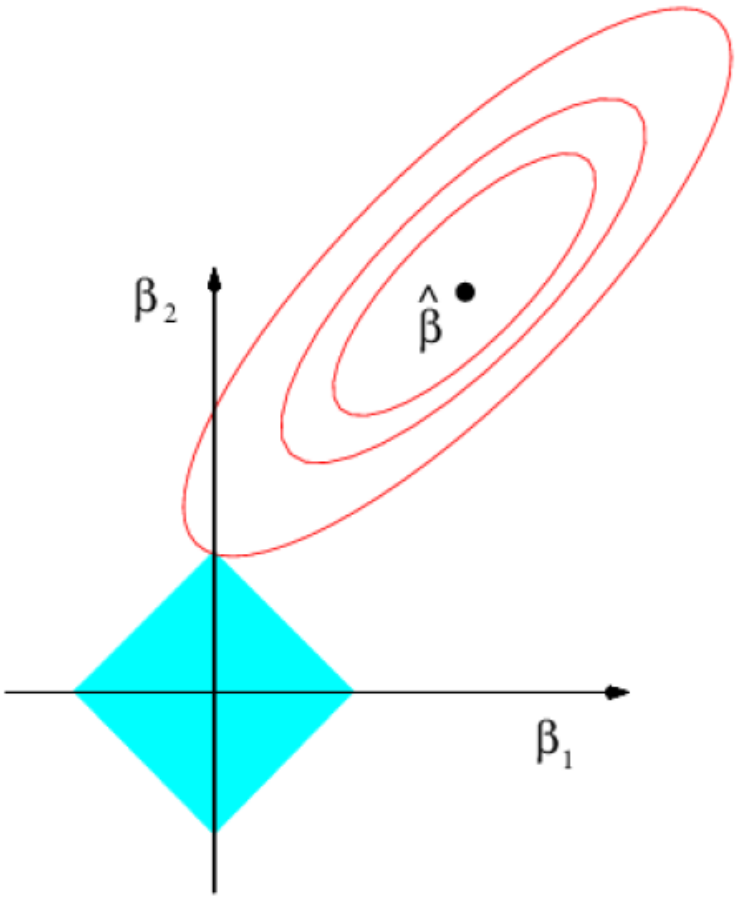
$$\hat{\beta}_{LASSO} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

- It is a *convex optimization* problem
- It is equivalent to solve a constrained optimization problem

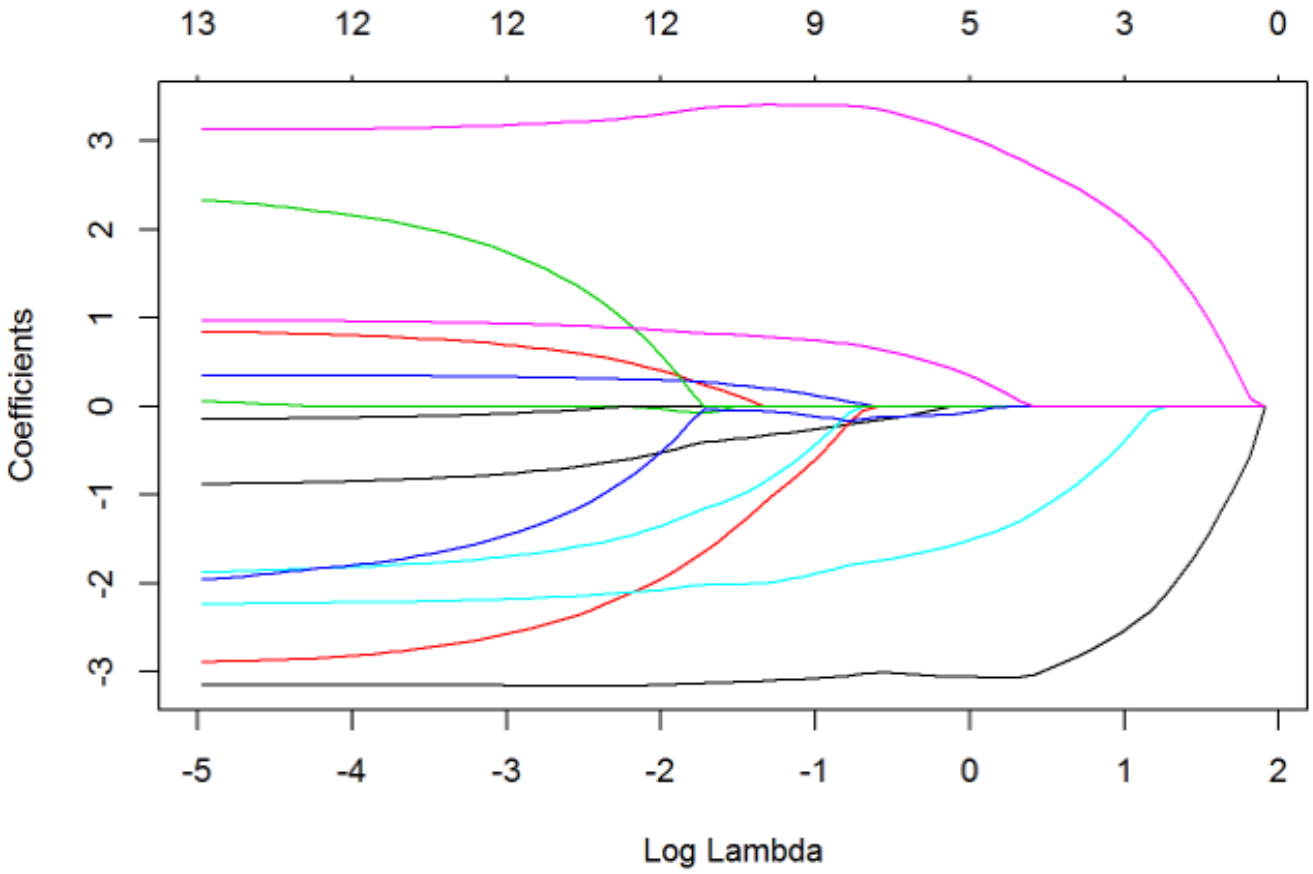
$$\min \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

$$\text{s.t.} \quad \sum_{j=1}^{p} |\beta_j| = a$$

An Illustration

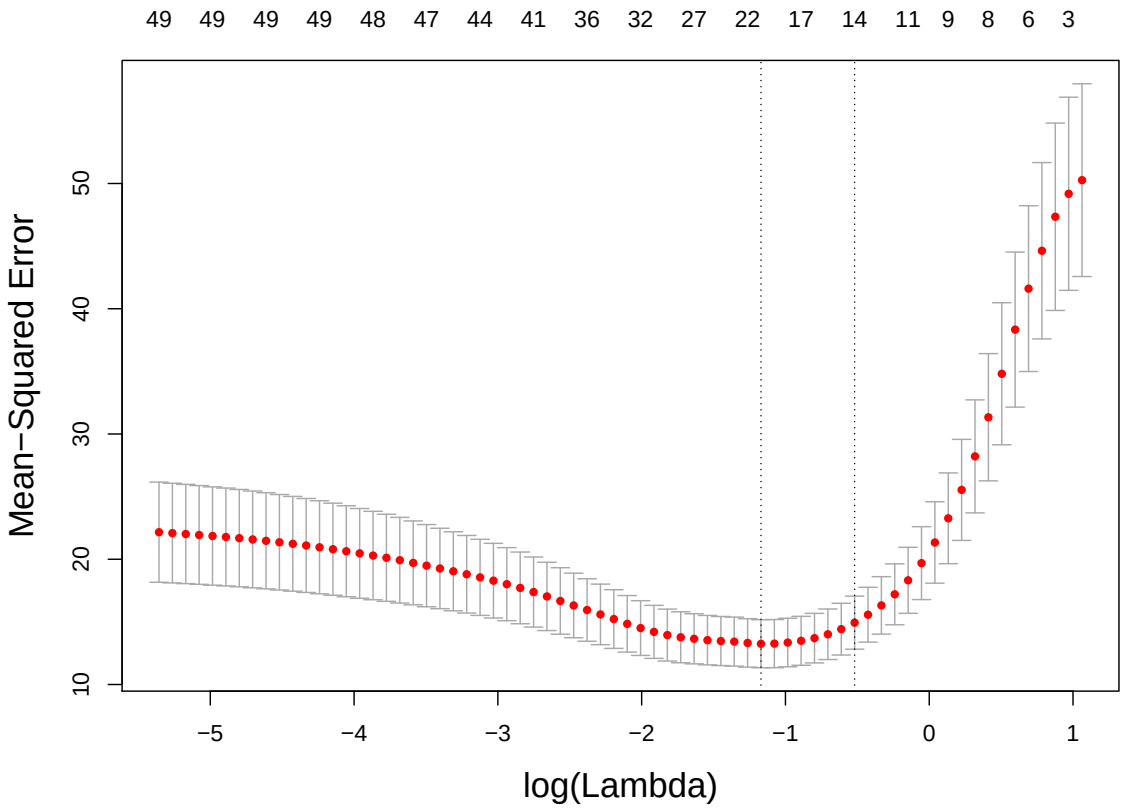


LASSO Regression Solution Path – Boston Housing Data



Tuning Parameter λ Selection

- λ controls the shrinkage level (different *lambda* associates with different estimated model)
- Cross-validation
  - In R, use the function `cv.glm()` in package `glmnet`



High-Dimensional Regression

- Number of predictor is very large (even larger than sample size)
- Ultra-high dimension $p \gg n$
- It is very common for gene expression and image data
- Sparsity assumption: only a few predictors are relevant
- OLS fails when $n < p$. Why?
- LASSO or similar methods provide sparse solution

Elastic Net Regression

- Introduced by Zou and Hastie (2005)
- Combination of Ridge and LASSO

$$\hat{\beta}_{EN} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^T \beta\right)^2 + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p} \beta_j^2$$

- Convex optimization
- Ridge and LASSO are special cases of Elastic Net
- It incorporates the advantages of both Ridge and LASSO
  - Ridge regression: lower variance; multicollinearity
  - LASSO: variable selection (selects at most n variables if $p > n$)

Some Notes

- We should standardize all predictors in shrinkage estimation. Why?
- Solution of Ridge regression is equivalent to the posterior of Bayesian estimates
- There are many other type of penalized estimators with different penalty functions that can perform variable selection.
  - Group Lasso (Yuan and Lin, 2006)
  - Adaptive-LASSO (Zou, 2006)
  - SCAD (Fan and Li, 2001)
  - MCP (Zhang, 2010)

Ridge Regression

- Recall least square. We solve the optimization

$$\hat{\beta}_{LS} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- Ridge regression solves a *(L_2) penalized least square*

$$\hat{\beta}_{Ridge} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

- λ is a tuning parameter, called shrinkage parameter
- Writing in matrix form, we can get the analytical solution

$$\hat{\beta}_{Ridge} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y} \quad \text{(Exercise: Show it!)}$$

Ridge Regression

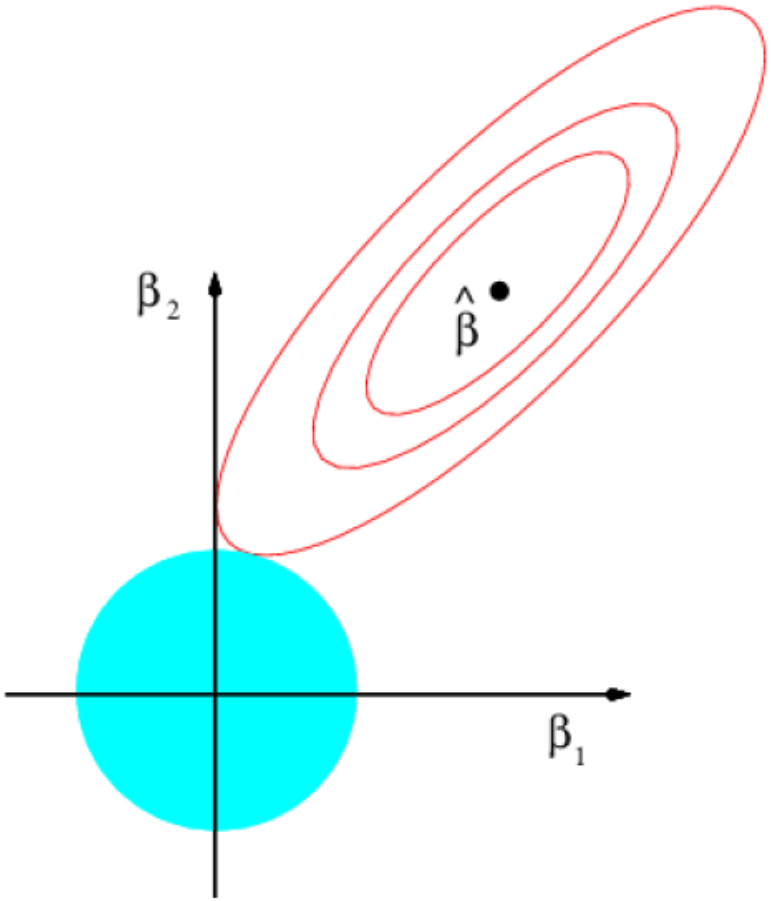
- It is equivalent to solve a constrained optimization problem

$$\min \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

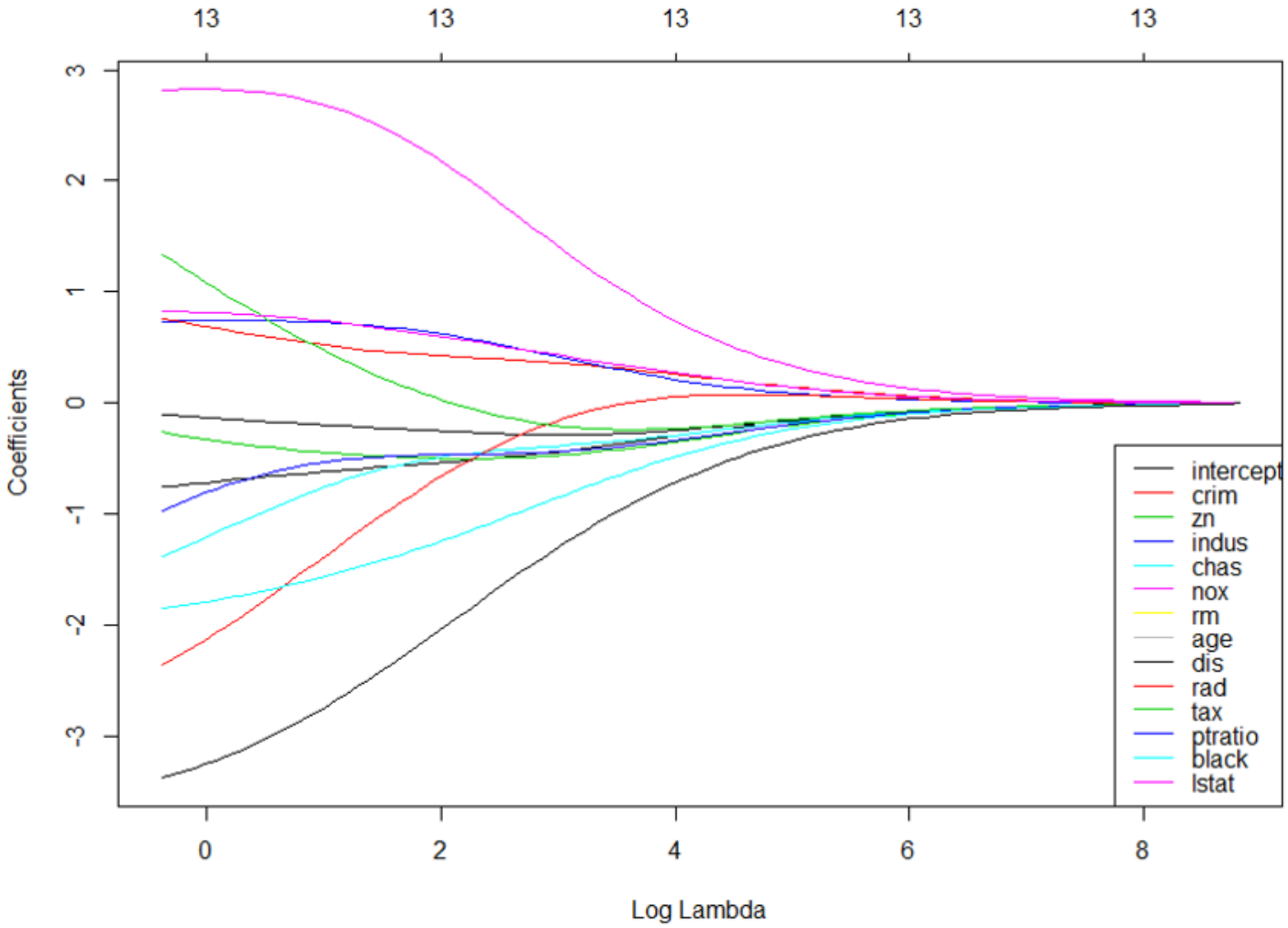
$$\text{s.t.} \quad \sum_{j=1}^{p} \beta_j^2 = a$$

- a corresponds to the tuning parameter λ

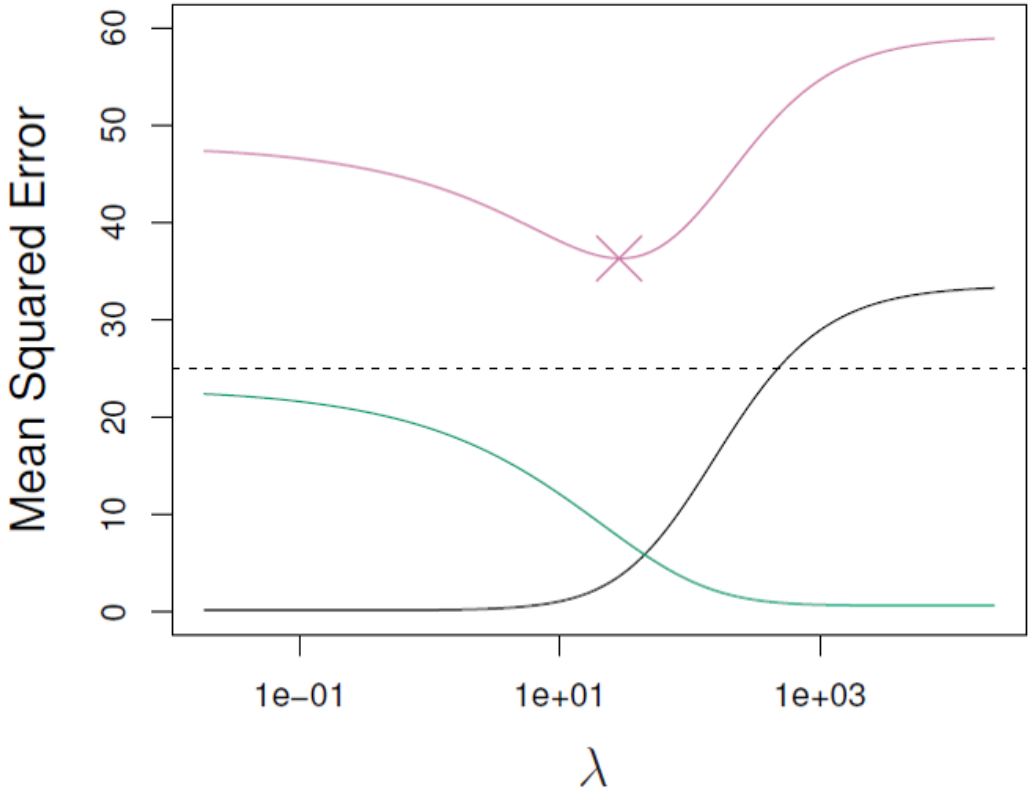
An Illustration



Ridge Regression Solution Path – Boston Housing Data



Bias-Variance Tradeoff



Simulated data with n=50 observations, p=45 predictors, all having nonzero coeficients. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set.